

Econ 272 / GSB 607
Spring 2023

Department of Economics
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PROBLEM SET V

DUE: SUNDAY MAY 11th, 2025, 6PM.

1. Here we will be using the Lalonde data, in the version posted by Rajeev Dehejia. There are two data files, one with the information for the treated individuals, and one with the information for the CPS comparison group. The variables are treatment indicator, age, education, black, hispanic, married, nodegree, earnings 1974, earnings 1975, and earnings 1978 (the outcome variable). For the PSID sample you need to convert the three earnings variables (earnings 1974, earnings 1975, and earnings 1978) to thousands of dollars by dividing them by 1000. Create two additional pretreatment variables / covariates, an indicator for earnings in 1974 being equal to zero (u'74, for unemployed), and an indicator for earnings in 1975 being equal to zero (u'75).
 - (a) Construct a table with the normalized differences for the ten covariates.
 - (b) Estimate the propensity score using a probit regression, just putting in the ten co- variates linearly.
 - (c) Estimate the average treatment effect for the treated, using weights. What are the weights, in terms of the propensity score? Report summary statistics for the weights for the control units, normalized so they average out to one. What are the five largest weights? What are the covariates for these five observations? Calculate the standard errors for this estimate using the bootstrap.
 - (d) Repeat (3) using logit estimates for the propensity score.
 - (e) Based on the probit propensity score estimates, construct five strata. Base these strata on the quintiles of the estimated propensity score distribution for the treated individuals. How many control units are there in each stratum?
 - (f) Calculate the normalized differences in each of the five strata, and then combine them by weighting them, with the weights proportional to the number of treated

units in each stratum. Are there any covariates that are still very unbalanced? Is there one stratum that is still very unbalanced? Which one is the most imbalanced?

- (g) Estimate the average effect within each stratum using linear regression, including a dummy for the treatment and including the ten covariates linearly, without interactions.
- (h) Estimate the average effect on the treated using the within-stratum estimates, with weights proportional to the number of treated in each stratum. Include the standard error, based on the bootstrap. How do they compare to the weighting estimates?
- (i) Are these estimates reasonable? Discuss some of the issues.

2. Theoretical Calculations

- (a) With

$$e(x) \equiv pr(W_i = 1 | X_i = x) \quad p \equiv pr(W_i = 1)$$

and $f_c(x)$ and $f_t(x)$ the densities of the covariates in the control and treated subpopulations, show that

$$V(e(X_i)) = p^2(1-p)^2 E \left[\left(\frac{f_t(X_i) - f_c(X_i)}{pf_t(X_i) + (1-p)f_c(X_i)} \right)^2 \right]$$

- (b) Also show that

$$V(e(X_i)) = p(1-p) \left(E[e(X_i) | W_i = 1] - E[e(X_i) | W_i = 0] \right)$$